

SAPTARSHI UPADHYAY

+91 8272918658 | Kolkata, West Bengal | upadhyay.saptarshi@gmail.com

linkedin.com/in/saptarshi-upadhyay | github.com/saptarshiUpadhyay2006 | upadhyaysaptarshi.xyz

EDUCATION

B.Tech in Information Technology, Jadavpur University at Kolkata, India

2024 – 2028

CGPA: **9.09**

Senior Secondary (XII), CBSE, Central Model School Barrackpore

2024

Percentage: **93.60%**

Secondary (X), ICSE, St. Xavier's Institution at Panihati

2022

Percentage: **96.80%**

SKILLS

Languages: C/C++, Java, JavaScript, TypeScript, Python, SQL

Frameworks & Tools: React, Next.js, React Native, Node.js, Express, MongoDB, Git, Docker, Kubernetes, AI/ML, NumPy, Pandas

Coursework: Data Structures & Algorithms, DBMS, Computer Networks, OOP, Software Engineering

EXPERIENCE

Software Engineer Intern

March 2026 - May 2026

Aarish AI Technologies Inc.

Remote

- Contributed to the **LORROS defence project** by resolving critical bugs and implementing feature corrections across backend **REST APIs** and database models.
- Engineered and deployed **AI-integrated mission-critical applications**, improving system reliability and achieving an estimated **5–8%** boost in processing throughput through query and payload optimization.
- Collaborated closely with senior engineers on testing, debugging, and code reviews, successfully streamlining system stability across **3** release cycles.

PROJECTS

Greenlens

Live | Demo | GitHub

- Architected a **microservice platform** using **Next.js**, **Node.js/Express**, **Python FastAPI**, and **MongoDB Atlas**, processing **1,000+** carbon-tracking records across **3** distinct microservices.
- Programmed an automated bill parser and ML engine using **Tesseract OCR**, **PyMuPDF**, and **XGBoost**, achieving an estimated **88%** OCR extraction accuracy and an **R² of 0.84** baseline prediction score across **4** invoice formats.
- Crafted an **interactive** analytics dashboard with **Tailwind CSS** and **Recharts** visualizing **6** core real-time telemetry and carbon simulation metrics.

HomeQuest

Live | Demo | GitHub

- Developed a full-stack rental marketplace using **Node.js**, **Express.js**, **EJS**, and **MongoDB Atlas** under an MVC architecture, supporting **500+** mock listings and **100+** registered users with dual authentication (Passport.js / Google OAuth).
- Assembled a secure payment and invoicing pipeline using **Razorpay SDK** with HMAC verification, processing **50+** sandbox test transactions at a **100%** success rate with server-side **PDFKit** receipt generation.
- Integrated geospatial and analytics features using **Mapbox GL JS** and **Cloudinary CDN**, optimizing MongoDB search queries across **500+** listings to return results in under **100ms**.

Binivex

Live | Demo | GitHub

- Constructed a real-time stock analytics platform using **Next.js**, **TypeScript**, **MongoDB**, **Finnhub API**, and **TradingView widgets**, tracking **30+** custom tickers with optimized **10-second** polling data refresh intervals.
- Automated AI stock intelligence reports and daily market digests using the **Google Gemini API** and **Inngest workflows**, generating **10+** automated reports daily across **3** background scheduled jobs.
- Implemented OAuth and email authentication using **Better Auth** and event-driven MongoDB hooks, onboarding **50+** test users with a **100%** auth success rate.

POSITIONS OF RESPONSIBILITY

Web Developer, ACM Student Chapter

Live

- Co-developed the **Synchronicity** official platform using **React** and **Tailwind CSS**, delivering optimized UI components for an active user base of **500+** student participants.

Development Coordinator, CodeClub JUSL

Live

- Spearheaded the development of the **CodeClub JUSL** official website and contributed to **Srijan 2026** (Kolkata's largest tech festival with **5,000+** attendees) as an active Tech Team member.
- Designed the high-concurrency merchandise platform, optimizing performance to seamlessly handle peak festival traffic across **3** days of events.

ACHIEVEMENTS

- Codeforces (Specialist):** Reached a peak standing of **1437** by solving **250+** algorithmic problems.
- LeetCode (Knight):** Peak rating of **1882** (Top **5%**) across **1,000+** problems.
- HackSprint:** Secured a position among the Top 8 national finalists at IEST Shibpur.
- East India Blockchain Summit (EIBS 2.0):** Advanced to the finals at IIT KGP.
- Double Slash 4.0 Hackathon (IEEE):** Reached the final round among competing teams.